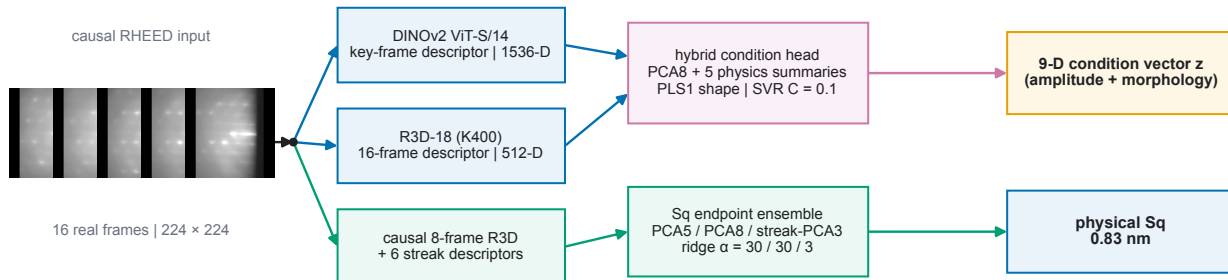


a

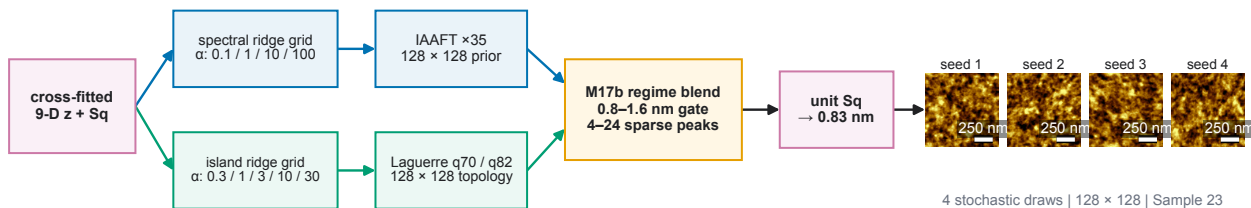
RHEED representation and condition prediction



strict outer LOO: every scaler, PCA and head is fitted on 26 growths

b

M17b topology-conditioned sparse-peak terrace generator



nonretrieval: measured AFM is unavailable at inference

c

Growth-level leave-one-out validation



27 outer folds | growth groups are the leakage boundary | one operator-invalid growth excluded before fitting